Ultracold ^{87}Rb in a vertical 1064 nm standing wave above a hollow-core fiber: physics background and simulation method

Semiclassical DSMC with band-resolved collisions

Project notes — `rb_lattice_dsmc`

Version 1, 23 October 2026

Abstract

These notes collect the physics needed to understand, implement and critically assess a direct-simulation Monte Carlo (DSMC) model of ultracold ^{87}Rb atoms ($T = 1$–$5\,\mu$K) held in a vertical, nearly counter-propagating 1064 nm standing wave whose beams are focused on the output facet of a hollow-core fiber (HCF). We review the optical dipole potential, the geometry of the two-beam field (wavefront curvature, Gouy phase, small crossing angle), band structure in a tilted 1D lattice, quasi-two-dimensional collisions and Bird's DSMC algorithm. We then describe the semiclassical model adopted here — quantized axial motion (band index), classical radial motion, band-resolved golden-rule collisions — together with the treatment of losses, of atoms falling into the fiber, and the definition of all observables stored by the simulation. The text is written for a PhD student in AMO physics; key references are given for each topic.

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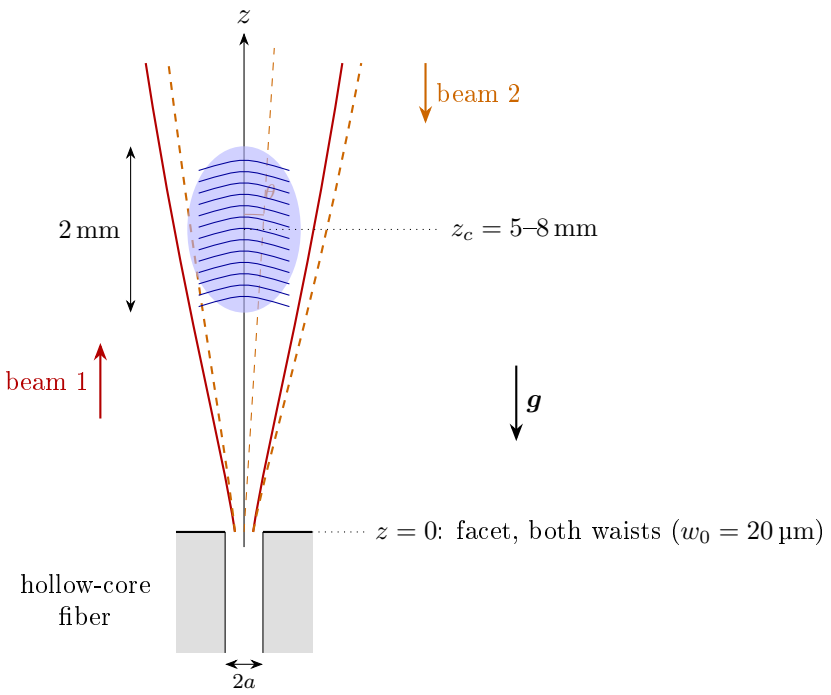


Figure 1: Geometry (not to scale). The lattice planes inside the cloud are curved (radius $\simeq z$) because both beams diverge from the same focus at the facet.

1 Overview of the problem

Cold atoms from an optical molasses are captured in a 1D optical lattice formed by two red-detuned beams at $\lambda = 1064\,\text{nm}$ (Fig. 1). Beam 1 exits a hollow-core fiber whose facet defines $z = 0$ and propagates upward ($+z$); beam 2 propagates downward, tilted by a small angle $\theta = 10^{-4}$–$10^{-2}\,\text{rad}$. Both beams have a waist $w_0 = 20\,\mu\text{m}$ located at the facet, and their axes cross there. The atom cloud (diameter 2 mm along z) is centred at $z_c = 5$–$8\,\text{mm}$, far beyond the Rayleigh range $z_R = \pi w_0^2/\lambda = 1.18\,\text{mm}$. Gravity points toward the fiber.

The questions the simulation must answer are: how many atoms stay trapped over $\sim 200\,\text{ms}$, through which channels the others are lost (radially, or by falling down the lattice onto/into the fiber), how the energy is distributed between kinetic and potential, axial and radial degrees of freedom, and how all this depends on two-body collisions. Table 1 anticipates the main scales; the rest of these notes explains where they come from.

2 Atom–light interaction: the optical dipole potential

2.1 Dipole potential and scattering rate

For a far-detuned field $\boldsymbol{E}(\boldsymbol{r},t) = \text{Re}[\boldsymbol{\mathcal{E}}(\boldsymbol{r})\text{e}^{-\text{i}\omega t}]$ the ground-state light shift is

$$U(\boldsymbol{r}) = -\frac{1}{4}\,\text{Re}\,\alpha(\omega)\,|\boldsymbol{\mathcal{E}}(\boldsymbol{r})|^2 = -\frac{\text{Re}\,\alpha(\omega)}{2\varepsilon_0 c}\,I(\boldsymbol{r}), \qquad I = \tfrac{1}{2}\varepsilon_0 c|\boldsymbol{\mathcal{E}}|^2, \tag{1}$$

and the photon scattering rate is $\Gamma_{\text{sc}} = \text{Im}\,\alpha\, I/(\hbar\varepsilon_0 c)$ [1]. For alkali atoms at 1064 nm both D lines contribute; the multilevel expression (including counter-rotating terms) is given in Ref. [1]. With linear polarization the vector light shift vanishes and the potential is the same for all hyperfine Zeeman states (the tensor part is negligible for the $5S_{1/2}$ ground state). We use the scalar dynamic polarizability $\alpha(1064\,\text{nm}) \simeq 687\,\text{a.u.}$ [2], with $\alpha_{\text{SI}} = 4\pi\varepsilon_0 a_0^3\,\alpha_{\text{a.u.}}$. Atoms are attracted to intensity maxima.

A useful rule of thumb is $\Gamma_{\rm sc} \approx |U|\,\Gamma/(\hbar|\Delta_{\rm eff}|)$; for the deepest lattice considered ($|U| \simeq 40\,\mu$K) this gives $\Gamma_{\rm sc} \lesssim 0.3\,{\rm s}^{-1}$, i.e. fewer than 0.06 photons per atom in 200 ms. Photon recoil heating in a lattice (band-changing Raman processes in the Lamb–Dicke regime) is analysed in Ref. [6]; it is an optional, small effect here.

2.2 Atomic data

mass	$m = 86.909\,{\rm u}$	[3]
s-wave scattering length, $\lvert F{=}1, m_F{=}{-}1\rangle$	$a = 100.4\,a_0$	[4]
polarizability at 1064 nm	$\alpha \simeq 687\,{\rm a.u.}$	[2]
recoil energy at 1064 nm	$E_{\rm R} = \hbar^2k^2/2m = h \times 2.03\,{\rm kHz} = k_{\rm B} \times 97\,{\rm nK}$	
three-body loss (thermal)	$K_3 \approx 4 \times 10^{-29}\,{\rm cm}^6\,{\rm s}^{-1}$	[5]

3 Geometry of the two-beam field

3.1 Gaussian beams

A TEM_{00} beam propagating along $+s$ with waist w_0 at $s = 0$ has [7]

$$\mathcal{E}(\rho, s) = \mathcal{E}_0 \frac{w_0}{w(s)} \exp\!\left[-\frac{\rho^2}{w(s)^2}\right] \exp\!\left[-{\rm i}\left(ks + \frac{k\rho^2}{2R(s)} - \zeta(s)\right)\right], \tag{2}$$

$$w(s) = w_0\sqrt{1 + s^2/z_R^2}, \qquad R(s) = s\left(1 + z_R^2/s^2\right), \qquad \zeta(s) = \arctan(s/z_R), \tag{3}$$

with $I(\rho, s) = \frac{2P}{\pi w(s)^2}{\rm e}^{-2\rho^2/w(s)^2}$. The expressions are valid for negative s (before the focus) with R and ζ odd functions of s. For $z_c = 5$–$8\,$mm we are in the far field: $w \simeq w_0 z/z_R = 87$–$137\,\mu$m, $R \simeq z$.

3.2 Interference of the two beams

Let beam 1 propagate along $+z$ (axis $\hat{\boldsymbol{z}}$ through the origin) and beam 2 along $-\hat{\boldsymbol{e}}_2$, with $\hat{\boldsymbol{e}}_2 = (\sin\theta, 0, \cos\theta)$, also through the origin. Beam 2 is converging toward the facet: its own propagation coordinate is $s_2 = -\boldsymbol{r}\cdot\hat{\boldsymbol{e}}_2 < 0$ and its transverse distance is $\rho_2^2 = r^2 - (\boldsymbol{r}\cdot\hat{\boldsymbol{e}}_2)^2$. Writing $\mathcal{E}_i = A_i{\rm e}^{-{\rm i}\varphi_i}$ and assuming parallel linear polarizations, the total intensity is

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Phi, \qquad \Phi(\boldsymbol{r}) = \varphi_1(\boldsymbol{r}) - \varphi_2(\boldsymbol{r}) + \phi_0, \tag{4}$$

where ϕ_0 is a constant (path-length) phase that merely translates the lattice. On the common axis ($\theta = 0$),

$$\Phi \simeq 2kz + \frac{k\rho^2}{R(z)} - 2\zeta(z). \tag{5}$$

Three consequences are important:

1. **Lattice spacing.** $|\nabla\Phi| \simeq 2k$, so the period is $d = \lambda/2 = 532\,$nm. The Gouy term changes the local period by a relative amount $\sim z_R/(kz^2) \sim 10^{-5}$: negligible, but included automatically if Φ is computed exactly.

2. **Curved planes.** Surfaces of constant Φ are $z \simeq z_j - \rho^2/(2R)$: domes of radius $R \simeq z$. Across one beam radius the plane bends by $w^2/2R \simeq 0.7\,\mu{\rm m} > d$. A product ansatz "Gaussian envelope $\times \cos^2 kz$" is therefore *wrong* for radial motion; the site surfaces must be followed.

3. **Crossing angle.** For small θ, $s_2 \simeq -(z + \theta x)$ and $\rho_2^2 \simeq (x - \theta z)^2 + y^2$. The planes tilt by $\theta/2$ and the axis of beam 2 is displaced by θz (0.5 µm for $\theta = 10^{-4}$, up to 80 µm for $\theta = 10^{-2}$). The intensity product $\sqrt{I_1 I_2}$, which sets the lattice modulation, is maximal on the bisector $x = \theta z/2$ and reduced by $\exp[-(\theta z)^2/(2w^2)]$ there (0.84 at $\theta = 10^{-2}$, $z = 7\,$mm).

3.3 Lattice depth, offset, and site surfaces

With $U_i = \alpha I_i/(2\varepsilon_0 c)$ the potential including gravity is

$$V(\boldsymbol{r}) = -U_1 - U_2 - 2\sqrt{U_1 U_2}\cos\Phi + mgz = V_{\text{off}}(\boldsymbol{r}) - U_{\text{lat}}(\boldsymbol{r})\cos^2\frac{\Phi}{2} + mgz, \tag{6}$$

$$U_{\text{lat}} = 4\sqrt{U_1 U_2}, \qquad V_{\text{off}} = -\left(\sqrt{U_1} - \sqrt{U_2}\right)^2. \tag{7}$$

For balanced, perfectly overlapped beams $V_{\text{off}} = 0$ and $U_{\text{lat}} = 4U_1$ (a factor 4 deeper than a single beam). Imbalance or displacement produces a non-modulated (running-wave) offset $V_{\text{off}} < 0$. Sites (antinodes) are the surfaces $\Phi(\boldsymbol{r}) = 2\pi j$, $j \in \mathbb{Z}$; we parametrize site j by its height $z_j(x, y)$, obtained by Newton iteration on $\Phi(x, y, z) = 2\pi j$ starting from $z_j(0, 0) \simeq jd$.

The gravitational energy per site is $mgd = k_{\text{B}} \times 55\,\text{nK}$, far below $U_{\text{lat}} = 6$–$42\,\mu\text{K}$: each site is an independent well, and gravity only matters (i) as a small tilt of the 1D lattice (Sec. 4.2), (ii) along the curved site surface, where $mgz_j(\rho) \simeq -mg\rho^2/2R$ is a weak radial anti-confinement (it reduces ω_r^2 by g/R, a $\lesssim 2\%$ effect), and (iii) for atoms that are no longer bound to a site.

4 Atoms in a tilted 1D lattice

4.1 Band structure: the Mathieu problem

Locally, along the normal to the site surfaces (coordinate $\xi \simeq z$), the potential is $-U_{\text{lat}}\cos^2(k\xi)$ (we drop the slowly varying V_{off}). With $x = k\xi$ and $s \equiv U_{\text{lat}}/E_{\text{R}}$ the Schrödinger equation becomes Mathieu's equation [8, 9]

$$\psi''(x) + [a - 2q\cos 2x]\,\psi(x) = 0, \qquad q = \frac{s}{4}, \qquad E = E_{\text{R}}\,a - \frac{U_{\text{lat}}}{2}, \tag{8}$$

(after the shift $x \to x + \pi/2$). Band $n = 0, 1, \ldots$ spans $E \in [E_{\text{R}}a_n(q) - U_{\text{lat}}/2,\ E_{\text{R}}b_{n+1}(q) - U_{\text{lat}}/2]$, where a_n, b_n are the Mathieu characteristic values (available in `scipy.special`). The gap between bands n and $n+1$ is $\Delta_n = E_{\text{R}}[a_{n+1}(q) - b_{n+1}(q)]$. In the deep-lattice limit

$$E_n \simeq -U_{\text{lat}} + \hbar\omega_z\left(n + \tfrac{1}{2}\right) - \frac{E_{\text{R}}}{4}\left(2n^2 + 2n + 1\right), \qquad \hbar\omega_z = 2\sqrt{s}\,E_{\text{R}}, \tag{9}$$

and the lowest Wannier functions are close to harmonic-oscillator states of width $l_z = \sqrt{\hbar/m\omega_z} = k^{-1}s^{-1/4}$. For our parameters $s = 65$–430, $\hbar\omega_z/k_{\text{B}} = 1.6$–$4\,\mu\text{K}$, $l_z \simeq 30$–$45\,\text{nm}$ and only 5–13 bands lie below the barrier. Because $\hbar\omega_z \sim k_{\text{B}}T$, the axial motion cannot be treated classically: this is the reason for the semiclassical model of Sec. 5. The Lamb–Dicke parameter $\eta_{\text{LD}} = kl_z/\sqrt{2} = s^{-1/4}/\sqrt{2} \approx 0.16$–$0.25$.

Two quantities derived from the bands are needed below. By the Hellmann–Feynman theorem, the mean lattice potential energy and kinetic energy in band n are

$$\langle V_{\text{lat}}\rangle_n = U_{\text{lat}}\frac{\partial E_n}{\partial U_{\text{lat}}} = s\frac{\mathrm{d}E_n}{\mathrm{d}s}, \qquad \langle T_z\rangle_n = E_n - \langle V_{\text{lat}}\rangle_n, \tag{10}$$

and $\partial E_n/\partial U_{\text{lat}} = -\langle\cos^2 k\xi\rangle_n \in (-1, 0)$: atoms in higher bands are less strongly attracted toward regions of larger U_{lat}.

4.2 Tilted lattice: Bloch oscillations, Wannier–Stark states and Landau–Zener tunnelling

Gravity adds a linear potential $F\xi$ with $F = mg$. In a tilted lattice the quasimomentum evolves as $\hbar\dot{q} = -F$, so atoms undergo Bloch oscillations at $\nu_B = Fd/h = 1.14\,\text{kHz}$ [11]. Equivalently the eigenstates are Wannier–Stark ladders spaced by Fd and localized over W_n/Fd sites, W_n being

the band width [13]. For deep bands $W_n \ll Fd$ and atoms are localized on a single site; for the topmost bound bands $W_n \sim 0.2$–$0.7\,\mu\mathrm{K}$ and the Bloch-oscillation amplitude spans a few to ~ 10 sites, which does not cause loss.

Loss along the lattice requires interband (Landau–Zener) tunnelling to the continuum. In the weak-binding form, the probability per Bloch period of crossing gap Δ_n is [12]

$$P_{\mathrm{LZ}} = \exp\!\left(-\frac{\pi\Delta_n^2}{4\hbar^2 k\, g}\right), \qquad \Gamma_{\mathrm{LZ}} \simeq \nu_B P_{\mathrm{LZ}}. \tag{11}$$

For all bound bands in our parameter range this gives $\Gamma_{\mathrm{LZ}} < 10^{-7}\,\mathrm{s}^{-1}$ (the gap above the highest bound band is still $\gtrsim 0.3\,\mu\mathrm{K}$). Tunnelling-induced loss is therefore negligible, and an atom leaves its site axially only if its energy is raised above the barrier, i.e. by a collision (or, optionally, photon scattering). We define a band as *bound* if its upper edge lies below the downhill barrier,

$$E_n^{\mathrm{top}}(s) < V_{\mathrm{off}} - \tfrac{1}{2}mgd, \tag{12}$$

and as *free* otherwise. Above-barrier Mathieu bands are wide with tiny gaps, so a free atom crosses them with $P_{\mathrm{LZ}} \simeq 1$: it accelerates down the lattice.

5 The semiclassical model

5.1 Born–Oppenheimer separation

The axial frequency exceeds the radial one by a factor $\omega_z/\omega_r \approx 300$–$600$. We therefore treat the axial motion adiabatically, as in the Born–Oppenheimer approximation: at each radial position $\boldsymbol{r}_\perp = (x, y)$ the atom occupies an eigenstate (band n) of the local 1D lattice with parameters $s(\boldsymbol{r}_\perp) = U_{\mathrm{lat}}(\boldsymbol{r}_\perp, z_j)/E_{\mathrm{R}}$ and $V_{\mathrm{off}}(\boldsymbol{r}_\perp)$, and the band index is an adiabatic invariant of the slow radial motion. The radial motion is classical ($k_{\mathrm{B}}T \gg \hbar\omega_r \approx 3$–$11\,\mathrm{nK}$) and governed by the band-dependent effective potential

$$V_n(\boldsymbol{r}_\perp; j) = V_{\mathrm{off}}(\boldsymbol{r}_\perp, z_j) + E_n\big(s(\boldsymbol{r}_\perp)\big) + mg\, z_j(\boldsymbol{r}_\perp), \tag{13}$$

with E_n measured from the lattice maximum. The radial force follows from Eq. (10):

$$\boldsymbol{F}_{\perp,n} = -\nabla_\perp V_{\mathrm{off}} + \langle\cos^2\rangle_n \nabla_\perp U_{\mathrm{lat}} - mg\nabla_\perp z_j. \tag{14}$$

In the harmonic limit this gives the band-dependent radial frequency $\omega_{r,n}^2 \simeq \omega_r^2\langle\cos^2\rangle_n$ (minus the curvature correction g/R), with $\omega_r^2 = 4U_{\mathrm{lat}}/(mw^2)$.

The state of a *bound* test particle is thus (j, n, x, y, v_x, v_y), and its total energy $E = \frac{1}{2}mv_\perp^2 + V_n(\boldsymbol{r}_\perp; j)$ is conserved between collisions. Non-adiabatic band changes driven by the radial motion are suppressed by the large frequency ratio.

Release at the beam edge. As an atom moves outward, $s(\boldsymbol{r}_\perp)$ decreases and band n may cease to satisfy Eq. (12). At that point the atom is converted into a free particle (next subsection). This is the natural semiclassical description of radial evaporation in a lattice: radially energetic atoms are released axially at the edge of the beam and fall.

5.2 Free (axially unbound) atoms

An atom in a band above the barrier moves across many sites per Bloch period. We propagate it classically in 3D in the lattice-averaged potential

$$\overline{V}(\boldsymbol{r}) = V_{\mathrm{off}} - \frac{U_{\mathrm{lat}}}{2} + mgz = -U_1(\boldsymbol{r}) - U_2(\boldsymbol{r}) + mgz, \tag{15}$$

i.e. the incoherent sum of the two beams. At conversion the atom is placed at $z = z_j(\boldsymbol{r}_\perp)$ with its radial velocity unchanged and an axial velocity $v_z = \pm\sqrt{2(E_n + U_{\rm lat}/2)/m}$ (random sign), which conserves energy on average. This approximation is good when the axial energy exceeds the barrier by more than a few band gaps, and is approximate just above threshold; gravity makes the subsequent motion downward in either case. The free atom falls toward the facet in $t_{\rm fall} \simeq \sqrt{2z_c/g} = 32$–$40\,\mathrm{ms}$, gaining $mgz_c/k_{\rm B} = 0.5$–$0.8\,\mathrm{mK}$, while being radially focused by the increasingly deep and narrow beams (the "optical funnel" exploited in Refs. [30, 28, 29]).

6 Collisions

6.1 Ultracold s-wave scattering

At μK temperatures only s-waves contribute. For identical bosons in the same internal state the cross section is $\sigma(k) = 8\pi a^2/(1 + k^2a^2)$, with $k = \mu g_{\rm rel}/\hbar$, $\mu = m/2$ [10]. The interaction can be replaced by the contact pseudopotential $V_{\rm int} = g\,\delta(\boldsymbol{r}_1 - \boldsymbol{r}_2)$, $g = 4\pi\hbar^2 a/m$. At $5\,\mu\mathrm{K}$ the mean relative wave number gives $ka \approx 0.2$, so the energy dependence of σ is a $\lesssim 4\%$ correction and the Born (pseudopotential) treatment is adequate; at $1\,\mu\mathrm{K}$, $ka \approx 0.08$. In a thermal 3D gas the mean collision rate per atom is $\Gamma_{\rm el} = n\sigma\bar{v}_{\rm rel}$, averaged over the density, with $\bar{v}_{\rm rel} = \sqrt{16k_{\rm B}T/\pi m}$.

6.2 Quasi-2D collisions in a lattice site

Here the axial size $l_z \approx 40\,\mathrm{nm}$ is much larger than $a \approx 5.3\,\mathrm{nm}$, so the collision itself is three-dimensional, but the axial motion before and after it is quantized: a collision can change the band indices $(n_1, n_2) \to (n_1', n_2')$, exchanging energy between the axial and radial degrees of freedom [14, 15]. For a pair in the same site, the golden-rule rate for the transition from $|n_1 n_2\rangle$ (relative radial momentum $\hbar\boldsymbol{k}$) into all final relative-momentum states of channel (n_1', n_2') is

$$\Gamma_{n_1 n_2 \to n_1' n_2'} = \frac{S\, g^2 \mu}{\hbar^3 A}\, \left|I_{n_1 n_2 n_1' n_2'}\right|^2\, \Theta\Big(E_{\rm rel} + E_{n_1} + E_{n_2} - E_{n_1'} - E_{n_2'}\Big)\,, \tag{16}$$

$$I_{n_1 n_2 n_1' n_2'} = \int \mathrm{d}\xi\; \phi_{n_1}(\xi)\phi_{n_2}(\xi)\phi_{n_1'}(\xi)\phi_{n_2'}(\xi), \tag{17}$$

where A is the quantization area, ϕ_n are the (real) Wannier functions, $E_{\rm rel} = \hbar^2 k^2/2\mu$, the 2D density of states $\mu A/(2\pi\hbar^2)$ has been used, and $S = 2$ is the bosonic symmetrization factor (the same factor that turns $4\pi a^2$ into $8\pi a^2$). Remarks:

- In 2D the density of states is energy independent: the rate does not depend on the radial relative speed except through channel opening thresholds (Θ function).
- For the ground band, $I_{0000} = 1/(\sqrt{2\pi}\, l_z)$ and $gI_{0000} = 2\sqrt{2\pi}\hbar^2 a/(m l_z)$ is the familiar quasi-2D coupling constant [14]. Beyond-Born corrections scale as $(a/l_z)\ln(\dots)$, a few percent here [15].
- Site parity symmetry implies $I = 0$ unless $n_1 + n_2 + n_1' + n_2'$ is even.
- When many bands are populated, summing Eq. (16) over final channels and averaging over thermal band populations reproduces the 3D result $n\sigma\bar{v}_{\rm rel}$. This is a validation test of the implementation.
- Final channels with one atom in a free band are included: this is the microscopic mechanism of axial evaporation.

6.3 Direct Simulation Monte Carlo

DSMC [16] solves the Boltzmann equation stochastically by alternating (i) collisionless propagation of N_{test} test particles, each representing $N_{\text{eff}} = N_0/N_{\text{test}}$ real atoms, and (ii) collisions between test particles that share a spatial cell, with probabilities that reproduce the local collision rate. Its application to cold-atom dynamics and its accuracy requirements (cell size small compared with the density variation and mean free path, time step small compared with the collision time, enough particles per cell) are discussed in Ref. [17]. In Bird's *no-time-counter* (NTC) scheme, a cell with N_c test particles and volume V_c is assigned

$$M_c = \left\lfloor \tfrac{1}{2} N_c (N_c - 1)\, N_{\text{eff}}\, (\sigma g)_{\max}\, \Delta t / V_c + \mathcal{U} \right\rfloor \tag{18}$$

candidate pairs ($\mathcal{U}$ uniform in $[0, 1)$), each accepted with probability $\sigma g/(\sigma g)_{\max}$; accepted pairs are scattered isotropically in the centre-of-mass frame. The scheme conserves energy and momentum exactly in each collision.

Adaptation to the lattice.

1. *Cells.* Only atoms in the same site can collide. One site holds too few test particles for good statistics, so a cell groups K adjacent sites with an (x, y) grid of area A_c. Neighbouring sites have identical distributions to within $\sim Kd/z \sim 10^{-3}$ for $K \lesssim 20$, so pairing a particle with one in a neighbouring site is statistically equivalent to pairing within the same site, provided the pair rate is divided by K: the effective cell "volume" is $V_c = KA_c$.

2. *Rate majorant.* The role of σg is played by $\Gamma_{\text{pair}} A = S g^2 \mu\, W/\hbar^3$ with $W = \sum_{\text{open } f} |I_{if}|^2$. Candidates are drawn with the majorant $W_{\max}$ and accepted with probability $W/W_{\max}$.

3. *Post-collision state.* Choose the final channel f with probability $|I_{if}|^2/W$; conserve the radial centre-of-mass velocity; set the new relative radial speed from energy conservation,

$$\tfrac{1}{2}\mu g_\perp'^2 = \tfrac{1}{2}\mu g_\perp^2 + E_{n_1} + E_{n_2} - E_{n_1'} - E_{n_2'}, \tag{19}$$

   and choose its direction uniformly in the plane (s-wave). Atoms that end in a free band are converted to free particles.

4. *Validity.* The method treats the gas as classical (Boltzmann statistics): Bose enhancement of the final states is neglected. Peak phase-space densities stay below $\sim 10^{-2}$ except in the most extreme corner of the scan (0.8 W, 5 mm, 1 µK, 10^6 atoms), where they approach 0.2 and the results should be read with care.

Free atoms are dilute and leave within tens of ms; their collisions are neglected.

7 Initial state

Atoms are loaded from a molasses at temperature T_0. The cloud envelope along z is a Gaussian with $1/e^2$ diameter 2 mm ($\sigma_{\text{cl}} = 0.5$ mm), which fixes the fraction of atoms in each site, $p_j \propto \exp[-(z_j - z_c)^2/2\sigma_{\text{cl}}^2]$. Within each site the atoms are distributed according to the potential, i.e. by the Boltzmann distribution in the semiclassical phase space:

$$P(n, \boldsymbol{r}_\perp, \boldsymbol{v}_\perp \,|\, j) \propto \exp\left[-\frac{\tfrac{1}{2} m v_\perp^2 + V_n(\boldsymbol{r}_\perp; j)}{k_{\text{B}} T_0} \right], \qquad \rho \le \rho_{\text{TR}}, \tag{20}$$

including bands above the barrier (initially free atoms) and radially unbound positions inside the trap region of radius ρ_{TR}. Initially unbound atoms are kept because at $\eta = U_{\text{lat}}/k_{\text{B}}T_0 \sim 1$–3 they are a large, physical part of the sample; they produce the fast initial loss transient. Sampling

proceeds as: site j from p_j; band n from the radial partition functions $Z_{n,j} = \int \mathrm{d}^2\boldsymbol{r}_\perp \, \mathrm{e}^{-V_n/k_\mathrm{B}T_0}$; position by rejection; velocity from a 2D Maxwell distribution. Note that this "equilibrium in the trap" initial condition differs from a sudden loading from the molasses (which would have the molasses density rather than the Boltzmann-compressed density [23]); the alternative can be added as an option.

8 Losses and the fiber

8.1 Trap region and loss channels

The trap region is $|z - z_c| \le z_\mathrm{TR}$ (default 1.25 mm) and $\rho \le \rho_\mathrm{TR} = 3w(z)$. An atom is *lost radially* when it leaves the region transversely, and *lost axially* when it falls below it. Lost-axially atoms are propagated further until they reach $z = 0$ (*in transit* until then), and classified there as hitting the facet ($\rho > a$) or entering the fiber core ($\rho < a$). Optional non-evaporative losses are one-body (background gas, rate γ_bg) and three-body, $\dot{n} = -K_3 n^3$, evaluated with the local density in each cell; the latter is $\lesssim 10^{-2}\,\mathrm{s}^{-1}$ here.

8.2 Entry into the fiber

The fiber core radius a is taken from optimal coupling of a Gaussian beam to the fundamental EH_{11} mode of a hollow capillary-like waveguide, $w_0 \simeq 0.64\,a$ [24, 25], giving $a \simeq 31\,\mu\mathrm{m}$ (a parameter). Inside the core the field is approximated by the fundamental mode, i.e. a Gaussian of constant width w_0; both beams propagate there and form a lattice of depth $4U_1(w_0) = 0.3$–$0.8\,\mathrm{mK}$. A falling atom has an axial energy of 0.5–0.8 mK and is not re-captured axially without dissipation. Its radial motion averages over the lattice, so we test radial confinement in the averaged potential $\overline{U}_f(\rho) = -\overline{U}_f \, \mathrm{e}^{-2\rho^2/w_0^2}$, $\overline{U}_f = U_1 + U_2$ at the facet. An atom entering at $(\rho, \boldsymbol{v}_\perp)$ is *radially bound in the fiber* if

$$\tfrac{1}{2}mv_\perp^2 - \overline{U}_f \, \mathrm{e}^{-2\rho^2/w_0^2} < -\overline{U}_f \, \mathrm{e}^{-2a^2/w_0^2}, \tag{21}$$

i.e. if its transverse turning point lies inside the core (atoms reaching the wall are lost). Experiments on guiding cold atoms in HCFs are reported in Refs. [26, 27, 28, 29, 30].

9 Observables

All per-atom averages are taken over atoms remaining in the trap region (and, where indicated, separately over bound atoms). Energies are measured from the trap minimum at z_c.

Kinetic energy
$E_\mathrm{kin} = \frac{1}{2}mv_\perp^2 + \langle T_z\rangle_n$ for bound atoms (Eq. (10)); $\frac{1}{2}mv^2$ for free atoms.

Potential energy
$E_\mathrm{pot} = E_\mathrm{tot} - E_\mathrm{kin}$, i.e. $V_n(\boldsymbol{r}_\perp; j) - \langle T_z\rangle_n$ for bound atoms and $\overline{V}(\boldsymbol{r})$ for free atoms.

Total energy
$E_\mathrm{tot} = E_\mathrm{kin} + E_\mathrm{pot}$; conserved atom by atom without collisions, and by every collision.

Virial
For a classical system in a potential V, the time average of $2K - \boldsymbol{r}\cdot\nabla V$ vanishes; for a stationary quantum state the same holds as an expectation value. We monitor $\mathcal{V} = \big\langle 2K_\perp - (\boldsymbol{r}_\perp - \boldsymbol{r}_\perp^{(0)})\cdot\nabla_\perp V_n\big\rangle + \big\langle 2\langle T_z\rangle_n - \langle \xi\,\partial_\xi V_\mathrm{lat}\rangle_n\big\rangle$, where $\boldsymbol{r}_\perp^{(0)}$ is the site's radial minimum. The axial term vanishes identically for band eigenstates; the radial term relaxes to zero in equilibrium. We also store the ratio $2\langle K_\perp\rangle / \langle(\boldsymbol{r}_\perp - \boldsymbol{r}_\perp^{(0)})\cdot\nabla_\perp V_n\rangle$ (equal to 1 in equilibrium; for a harmonic potential it also equals $2\langle K\rangle/2\langle V\rangle$).

Temperatures
$T_\perp = m\langle v_\perp^2\rangle/(2k_\mathrm{B})$ (after removing the mean velocity). $T_\parallel$ is defined by matching the mean band energy to its thermal value, $\sum_n E_n \mathrm{e}^{-E_n/k_\mathrm{B}T_\parallel} / \sum_n \mathrm{e}^{-E_n/k_\mathrm{B}T_\parallel} = \langle E_n\rangle$, over bound bands. A kinetic temperature $T_\mathrm{kin} = 2\langle E_\mathrm{kin}\rangle/(3k_\mathrm{B})$ is stored for comparison with the classical code. $T_\perp \neq T_\parallel$ signals incomplete cross-dimensional thermalization [22].

Fractions
(relative to N_0) remaining in the region; bound; unbound-in-region; lost radially; in transit; hit facet; entered fiber; radially bound in fiber; other losses.

Decay time
$N_\mathrm{reg}(t)$ is fitted for $t > t_\mathrm{fit}$ (after the initial transient) with (i) $N_\infty + (N_0' - N_\infty)\mathrm{e}^{-t/\tau}$ and (ii) the 1-body+2-body model $\dot N = -N/\tau_1 - \beta' N^2$. Without collisions $N(t)$ plateaus after the transient (apart from residual slow radial escape), so τ characterizes the transient only; the comparison with the collisional run isolates evaporation.

Collision probability
Each test particle carries a collision counter. We store the instantaneous elastic rate per atom $\Gamma_\mathrm{el}(t) = 2N_\mathrm{coll}/(N\Delta t)$, the cumulative probability $P_{\geq 1}(t)$ that an atom has collided at least once, the mean number of collisions per atom, the mean number before loss, and the fraction of band-changing collisions.

10 Evaporation: what to expect

Kinetic theory of evaporation [19, 20] gives, for a truncation parameter $\eta = U/k_\mathrm{B}T \gtrsim 4$, an evaporation rate per atom $\Gamma_\mathrm{ev} \sim \Gamma_\mathrm{el}\,\eta\,\mathrm{e}^{-\eta}$ (with geometry-dependent prefactors), and a temperature that drifts toward $T \simeq U/\eta_\infty$ with $\eta_\infty \approx 8$–10 in the absence of other losses. About 2.7 collisions per atom are needed for thermalization in a harmonic trap [22]. Two features specific to this system:

- The escape thresholds are anisotropic: radial escape needs radial energy $\sim U_\mathrm{lat}$ (minus the band energy), axial escape requires a collision that deposits $\sim U_\mathrm{lat}$ into a single axial degree of freedom, constrained by the band spectrum and the overlap integrals. Evaporation in reduced dimensions is less efficient than in 3D, and slow ergodic mixing matters [21].
- Over 200 ms with $\Gamma_\mathrm{el} \approx 0.01$–$180\,\mathrm{s}^{-1}$, the scan spans the collisionless, the few-collision, and the fully thermalizing regimes. The total loss is the sum of the initial transient (unbound atoms, independent of collisions) and evaporation; only the latter depends on N_0.

For the low-η regime and non-ergodic traps, DSMC avoids the truncated-Boltzmann and ergodicity assumptions made in rate-equation models such as Ref. [18].

11 Numerical implementation

11.1 Pre-computation

1. Lookup tables vs $s = U_\mathrm{lat}/E_\mathrm{R}$ (log-spaced grid, $s \in [1, 10^3]$, cubic interpolation): band edges and centres $E_n(s)$, $\mathrm{d}E_n/\mathrm{d}s$, Wannier functions ϕ_n (from a plane-wave or finite-difference diagonalization), overlap integrals $I_{n_1 n_2 n_3 n_4}(s)$ for $n \leq n_\mathrm{max}$, and the index of the highest bound band.
2. Field model: $U_{1,2}(\boldsymbol{r})$, $\Phi(\boldsymbol{r})$ and their gradients from Eq. (2); site surfaces $z_j(\boldsymbol{r}_\perp)$ by Newton iteration.

11.2 Time step

Bound atoms: velocity Verlet [31] in V_n, with $\Delta t \approx T_r/100$–$T_r/200$ ($T_r = 4$–$18\,\mathrm{ms}$), i.e. $\Delta t \approx 20$–$50\,\mu\mathrm{s}$ and $\lesssim 10^4$ steps for 200 ms. This is the gain of the semiclassical model: a classical treatment of the axial motion would require $\Delta t \lesssim 1\,\mu\mathrm{s}$. Free atoms: velocity Verlet in $\overline{V}$ with sub-steps (radial frequencies reach $\sim 2\,\mathrm{kHz}$ near the facet). Collisions: $\Delta t\,\Gamma_{\mathrm{el}} \ll 1$, satisfied by a wide margin.

11.3 One time step

1. **Bound atoms**: half kick, drift in (x, y), update $s(\boldsymbol{r}_\perp)$ and check Eq. (12) (release → free), new force, half kick; check $\rho > \rho_{\mathrm{TR}}$ (lost radially).
2. **Free atoms**: sub-stepped Verlet in $\overline{V}$; classify exits (radial, below region → in transit; $z \le 0$ → facet or fiber, with the radial-boundness test).
3. **Collisions** (if enabled): build cells, NTC candidate selection with $W_{\max}$, acceptance, channel choice, update of $(n, \boldsymbol{v}_\perp)$, conversion of atoms promoted to free bands, increment of collision counters.
4. **Optional losses**: one-body, three-body, photon scattering.
5. **Record** observables every N_{rec} steps.

11.4 Validation

1. Forces vs finite-difference gradients; band energies vs Eq. (9) and tabulated Mathieu values.
2. Collisionless run: energy conserved per atom; $N(t)$ constant after the transient.
3. Detailed balance: starting from the Boltzmann distribution in a deep lattice, collisions must leave band populations and $T_\perp$ stationary.
4. Collision rate in a uniform quasi-2D gas vs Eq. (16); 3D limit with many bands populated.
5. Convergence in Δt, N_{test}, A_c and K; seed-to-seed scatter as error bar.

11.5 Validation results obtained so far

The test suite (`tests/`) implements the checks above. Representative results:

- Two-beam field: Φ agrees with Eq. (5) to 10^{-6} rad; site sag equals $-\rho^2/2R$ to 0.2%; at $\theta = 10^{-2}$ the depth on the bisector is reduced by $\exp[-(\theta z)^2/(2w^2)]$ and the planes tilt by $\tan(\theta/2)$.
- Bands: edges agree with Mathieu characteristic values to 10^{-6}; the ground-band radial frequency of V_0 agrees with $\omega_r\sqrt{\langle\cos^2\rangle_0}$ (corrected by g/R) to 0.2%.
- Initial sampling: band populations, $\langle\rho^2\rangle$, bound fraction and velocity variances agree with direct quadrature of the Boltzmann distribution within statistical errors; no sampler violations.
- Collisionless run (0.5 W, $z_c = 7\,\mathrm{mm}$, $T_0 = 4\,\mu\mathrm{K}$, 10^4 test particles, 200 ms): per-atom energy conserved to $3 \times 10^{-4}\,k_B T_0$, decreasing as Δt^2; the bound fraction equals the escape-energy prediction at all times; virial ratio 1.000 ± 0.002; all fractions converged in Δt to one test particle in 10^4.
- Golden rule: summed over channels and thermally averaged, the quasi-2D rate exceeds the 3D rate $n\,8\pi a^2 \bar{v}_{\mathrm{rel}}$ by 24%, 8%, 2.6% at $k_B T/\hbar\omega_z = 1, 2, 4$, approaching it as $k_B T/\hbar\omega_z \to \infty$ (this fixes $S = 2$ and the units of I).

- DSMC: in a uniform quasi-2D gas the simulated elastic rate equals the golden-rule value (0.984 ± 0.017); energy, radial momentum and the parity of $\sum n$ are conserved in every collision; a gas prepared with $T_\perp = 8\,\mu$K and all atoms in band 0 relaxes to $T_\perp = T_\parallel = 4.65\,\mu$K after ≈ 12 collisions per atom.

12 Reference scales

Table 1: Scales at the cloud centre for balanced beams and $\theta \to 0$. η is given for $T_0 = 1$ and $5\,\mu$K. $\Gamma_{\rm el}$ is the peak elastic rate for $N_0 = 10^4$ / 10^6 at $T_0 = 1\,\mu$K (thermal, harmonic estimate).

P (W)	z_c (mm)	w (µm)	$U_{\rm lat}$ (µK)	s	bound bands	ν_z (kHz)	ν_r (Hz)	η
0.3	5	87	15.6	160	8	51	141	15.6 / 3.1
0.3	8	137	6.3	65	5	33	57	6.3 / 1.3
0.8	5	87	41.6	428	13	84	231	41.6 / 8.3
0.8	8	137	16.8	173	9	53	93	16.8 / 3.4

P (W)	z_c (mm)	$\Gamma_{\rm el}$, $N_0 = 10^4$ ($\rm s^{-1}$)	$\Gamma_{\rm el}$, $N_0 = 10^6$ ($\rm s^{-1}$)
0.3	5	0.4	41
0.3	8	0.04	4.2
0.8	5	1.8	177
0.8	8	0.18	18

Further scales: $d = 532\,$nm (≈ 3760 sites in 2 mm); $mgd/k_{\rm B} = 55\,$nK, $\nu_B = 1.14\,$kHz; atoms per site at the cloud centre ≈ 4 ($N_0 = 10^4$) to ≈ 420 ($N_0 = 10^6$); free-fall time to the facet 32–40 ms; lattice depth at the facet 0.3–0.8 mK.

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